SHORT · ACID TRANSACTIONS

What are ACID transactions?

A transaction is a change to your data that counts as one unit. ACID is four promises that keep that change safe.

Key points

- Atomicity means all or nothing.
- Consistency means the table keeps its rules.
- Isolation means readers never see a half-finished write.
- Durability means a committed change stays saved.
- Delta tables give you ACID by default.

Tiny example

Create a small orders table, insert two rows, then update one order to shipped. In Databricks Free Edition, click New, then Notebook, and run each cell with Shift + Enter. No live output is recorded here.

```
CREATE TABLE orders (id INT, status STRING);

INSERT INTO orders VALUES (1, 'new'), (2, 'new');

UPDATE orders SET status = 'shipped' WHERE id = 1;
```

Misconception check

> ACID does not mean a change can never fail. If that update failed partway through, no partial change would be kept. Anyone reading the table sees the old version or the new one, never a mix.

Remember

With ACID, every change lands completely, or not at all.

The Short says you can try this in a notebook in Databricks Free Edition.